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Intro To C Programming

PSI Phase 1 Check-In

7/25/2026

Part 1:

```
psi> (+ 3 (* 4 5))
```

23

```
psi> (- 14 7)
```

7

```
psi> (* -1 -1)
```

1

```
psi> --1
```

-

```
psi> (/ 100 9)
```

11

```
psi> (= 4 +4)
```

#t

```
psi> (= 25 (* 5 5))
```

#t

```
psi> (= 7 8)
```

#f

```
psi> (= 7 (/ 1 0))
```

error: cannot divide by 0!

```
psi> (+ (* (+ 3 4) (+ 5 6)) (* (- 7 2) (/ 8 1)))
```

117

```
psi>()
```

```
()  
psi>  
Syntax error, not expecting a )...  
psi>(/ 8 -3)  
-3  
psi>(* 65536 65536)  
4294967296  
psi>( + 3 #t)  
$type error: got a bool when expecting a number...  
psi>(* 0 (/ 1 0))  
error: cannot divide by 0!  
psi>( + (* 3 #f) (/ 1 0))  
$type error: got a bool when expecting a number...  
psi>(3 4 5)  
error: inapplicable head  
psi>(quit 1)  
$error: arity-error, quit cannot have > 0 arguments...  
psi>(quit)  
0
```

Part 2:

I used linked lists to store the PSI lists. I used linked lists since they are non-contiguous in memory and they live in the heap. I am unsure of any better data structure to use. A worse data structure for a list would be a dynamically allocated array since it lives in the stack.

Part 3:

I am very confident that there are no memory errors, both illegal accesses and memory leaks. I used Valgrind to verify.